Lab Exercise 19- ClusterIP in Service in Kubernetes

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Batch-2

A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels) Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1 kind:
Deployment metadata:

  name: nginx-app      spec:

  replicas: 2
selector:    matchLabels:

      app: nginx   # MUST
match template labels   template:    metadata:
labels:
      app: nginx  # MUST match service selector
spec:      containers:      - name: nginx

image: nginx        ports:

      - containerPort: 80
```

Save file.

kubectl apply -f nginx-deployment.yaml

```
D:\kubs-misha>kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app created
```

Verify:

kubectl get pods --show-labels

```
D:\kubs-misha>kubectl get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE   LABELS
nginx-app-7769f8f85b-t59fr    1/1     Running   0          14s   app=nginx,pod-template-hash=7769f8f85b
nginx-app-7769f8f85b-zm4lh    1/1     Running   0          14s   app=nginx,pod-template-hash=7769f8f85b
```

You should see:

app=nginx

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion:   v1
kind:     Service
metadata:
  name: nginx-service spec:
  type: ClusterIP   selector:
    app: nginx   # MUST match pod label
ports:   - port: 80     targetPort: 80
```

Apply:

kubectl apply -f nginx-service.yaml

```
D:\kubs-misha>kubectl apply -f nginx-service.yaml
service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

kubectl get svc

```
D:\kubs-misha>kubectl get svc
NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
kubernetes      ClusterIP   10.96.0.1       <none>        443/TCP   3m41s
nginx-service   ClusterIP   10.96.154.144   <none>        80/TCP    8s
```

You should see:

TYPE CLUSTER-IP ClusterIP 10.x.x.x

MOST IMPORTANT CHECK

kubectl get endpoints nginx-service

```
D:\kubs-misha>kubectl get endpoints nginx-service
NAME            ENDPOINTS                     AGE
nginx-service   10.244.0.5:80,10.244.0.6:80   43s
```

Now you should see something like:

10.244.x.x:80, 10.244.x.x:80

Step 4: Test From Inside Cluster Run

test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

```
D:\kubs-misha>kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs, including credentials and sensitive inform
ation passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
```

You should see:

```
Welcome to nginx!
```

Exit:

```
exit
```

Step 5: Test From Outside (It Will Fail) From your local machine:

```
D:\kubs-misha>curl http://10.96.154.144
curl: (28) Failed to connect to 10.96.154.144 port 80 after 21075 ms: Could not connect to server

D:\kubs-misha>
```

curl http://<CLUSTER-IP>

It will NOT work.

That proves:

ClusterIP works only inside cluster.